



GIS KPI Dashboard — Application & Performance Workflow

The end-to-end operational flow of the GIS Team KPI Performance Dashboard — from account setup and KPI activity capture to evidence verification, period approval, deterministic scoring and auditable reporting.

DOCUMENT

Operational Workflow Guide

PLATFORM

GIS KPI Dashboard

SURFACES

Management Dashboard · Employee Workspace · Reports & Exports

VERSION

2.0

CLASSIFICATION

Internal Use

● MANAGEMENT DASHBOARD

Team Performance & Oversight

gis-kpi.vercel.app/overview

Executive overview, team scorecards, analytics, review queue, data quality and the audit log — open in a browser and updated live.

● EMPLOYEE WORKSPACE

Activity Capture & Evidence

gis-kpi.vercel.app/activities

Each employee captures activity against their five KPIs, attaches evidence and tracks their own attainment — same sign-in, scoped to self.

01 Platform overview & architecture

02 Roles & access model

03 Account setup & employee linking

04 KPI catalogue & source of truth

05 Activity capture & measurement

06 Evidence & verification

07 Review & period approval

08 Deterministic scoring engine

09 Analytics & team oversight

10 Reports, exports & AI narrative

11 Data quality, audit & notifications

One dashboard, every score reproducible

The GIS KPI Dashboard digitises how the GIS Team (Technical Services) plans, measures and proves performance. Every employee carries five core KPIs from the approved 2026 performance workbook plus four shared non-core KPIs — 100 weighted points in all; every reported result is backed by captured activity and reviewable evidence; and every official score is frozen as a reproducible snapshot an auditor can trace back to its inputs.

One web application serves three ways of working, all against a single backend:

- **The Management Dashboard** — for managers, admins and executives to monitor team performance, run the review queue, resolve data-quality issues and export reports.
- **The Employee Workspace** — where each employee captures KPI activity with mode-tailored forms, attaches evidence and follows their own attainment.
- **Reports & Exports** — deterministic Excel and PDF outputs, with an optional AI-written narrative that explains — but never produces — the numbers.

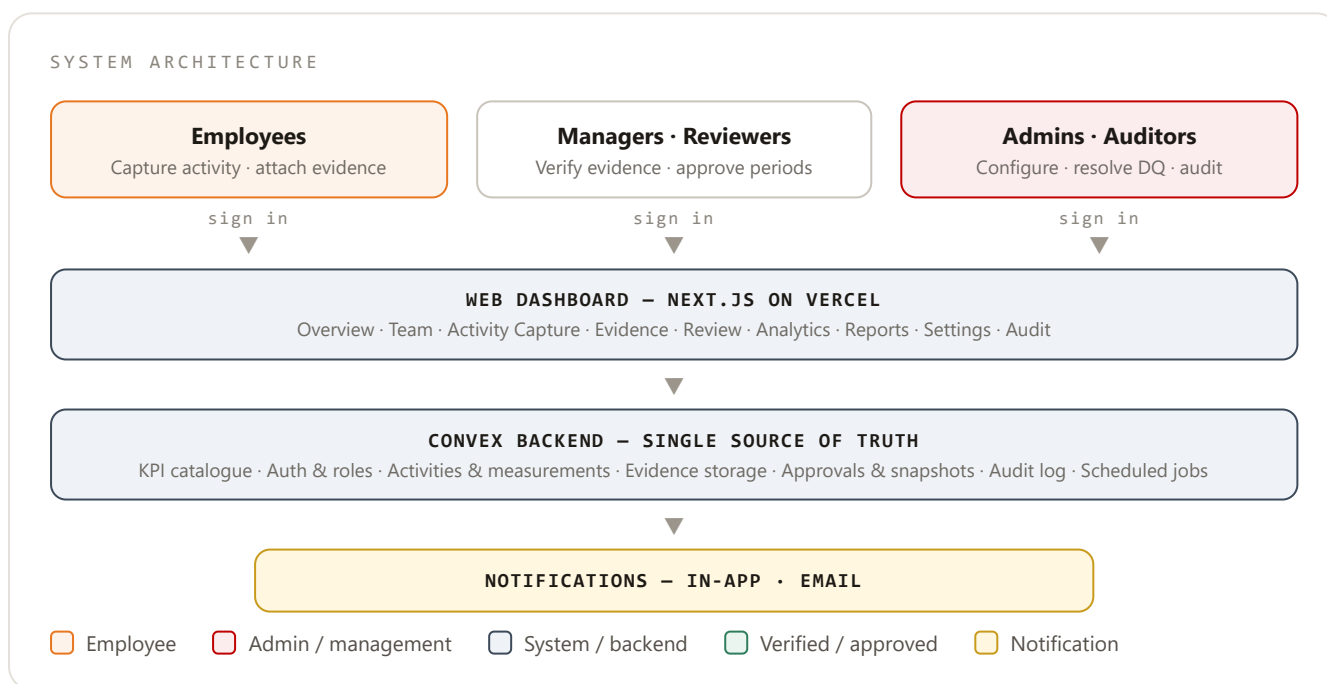


FIG. 1 — Employees, reviewers and administrators reach one backend through one dashboard. Every write is role-checked server-side, and the backend fans out review and approval alerts in-app and by email.

Seven roles, least-privilege by design

Access is granted by application role — kept strictly separate from an employee's *job role* on the roster.

Everyone signs in with their own email and password; what they can see and do follows from the roles granted to them. Authorization is enforced **server-side in every backend function** — hidden buttons are never the security boundary — and managers and reviewers can additionally be **scoped** to specific employees, job roles or locations.

Role	Dashboard	Capture activity	Verify & approve	Reports & exports	User management
System Admin	Full	Yes	Yes	Yes	Yes
KPI Admin	Full	Yes	Verify only	Yes	No
Manager	Scoped	Yes	Yes	Yes	No
Reviewer	Scoped	No	Verify	No	No
Employee	Self	Yes — own KPIs	No	No	No
Executive Viewer	Read-only	No	No	Approved only	No
Auditor	Audit views	No	No	Metadata	No

- **System Admin** — configures the organization: people, employee IDs, app roles, locations, managers, thresholds and periods. The only role that manages user accounts, and the only one that can lock a period or override with reason.
- **KPI Admin (GIS Team Admin)** — owns the KPI catalogue: templates, assignments, targets, weights, formulas, evidence rules, imports and the data-quality queue. Cannot manage user accounts.
- **Manager (GIS Lead)** — sees team performance within scope, verifies evidence, recommends and approves scores, and approves periods for their people.
- **Reviewer** — verifies evidence and recommends scores for the employees in their scope; never approves periods.
- **Employee** — captures activity against their own five KPIs, attaches evidence and follows their own score. Sees only their own restricted evidence.
- **Executive Viewer** — read-only, and only *approved* results — provisional numbers are never presented as final.
- **Auditor** — reads config history, score lineage and the audit log; sees evidence metadata without opening restricted files.

Every account is granted, linked, then trusted

Anyone can create an account, but a new account can do nothing until an administrator grants it a role.

Employees are additionally **linked to their roster record**, which is what puts their KPIs in front of them.

Unlinking is a clean break: the account's captured activity and evidence are **wiped by default** (an explicit keep-data option exists for genuine account moves), so a relink always starts clean — and either path is audit-logged.

1

Sign up

Email + password at /signin. The account exists but holds no role.

2

Awaiting access

The profile shows *'Awaiting access — no role granted yet.'* No data is visible.

3

Admin grants & links

In Users & Organization, a System Admin grants roles and links the account to its employee record.

4

Workspace live

The employee's KPIs appear in Activity Capture and on their Individual Performance page.

First-run bootstrap. On a fresh deployment, the first signed-in user can claim **System Admin** from the overview — the claim works only while no admin exists, and can be locked to a specific address with ADMIN_BOOTSTRAP_EMAIL. After that, every role passes through an admin's hands.

Profile & identity

Each user manages their own profile: display name, a profile photo (camera capture or upload, stored in Convex file storage), granted application roles, and — once linked — their designated roster job role, location and employee ID. Profile changes apply instantly across the dashboard and are audit-logged.

Scoped management

Managers and reviewers are granted their role **with a scope** — by employee, job role or location. A reviewer scoped to `Geo Database Specialist` sees exactly those people in the review queue and nothing more; team views outside a manager's scope simply do not render. Scope is evaluated on the server on every query.

Separation of duties. Application roles (what you may do in the app) are deliberately separate from roster job roles (what your KPIs measure). Granting someone Reviewer never changes their own KPI targets, and being a GIS Lead on the roster grants no dashboard power by itself.

From the approved workbook to a living catalogue

The catalogue is seeded from the approved 2026 performance workbook **GIS 2026 P0 Settings** — the same sheet management signed off. Every row is imported **verbatim** into a preserved source layer, then mapped to a canonical definition the scoring engine can compute. The original wording is never discarded: any screen that shows a canonicalised value can also show the source cell it came from.

75

Core KPI assignments

15

GIS Team employees

4

Job-role templates

100

Points per person — 80
core + 20 non-core

Each employee carries **five core KPIs** from their job-role template — Geographic Information Systems Lead, Geo Database Specialist, and the other 2026 role templates — plus **four shared non-core KPIs**, each with a target, a measurement mode, a weight in points and an evidence requirement.

Core 80 + non-core 20 = 100. The workbook's five core weights total **80 points**, preserved verbatim and never silently rebased. The remaining 20 points were closed by an explicit, audit-logged decision: four shared non-core KPIs adopted from the 2025 workbook — **safety hazard reporting** ($\geq 1/\text{month}$), **compliance recertification** (vs 80%, annual), **internal customer satisfaction** (vs 85%, annual) and **training hours** (20 hrs/yr; 10 for GL4+) — at 5 points each, for every role. Core and non-core are labelled apart on every scorecard.

Versioned configuration

KPI definitions are versioned: changing a target, weight, formula or threshold creates a new version with an effective period, and past measurements keep pointing at the version they were scored under. Assignments carry an immutable snapshot of what was agreed, so a mid-year configuration change can never quietly rewrite an old score.

Nothing is silently corrected. Where the workbook is ambiguous — a blank frequency, a location spelled two ways, a unit mismatch — the anomaly becomes a **data-quality issue** for an admin to resolve (§11). And where a source encoding punished honest reporting, the fix was an explicit migration: QA data quality is scored as a **coverage ratio** (errors corrected $\div$ identified), the commercial-maintenance KPI as a **monthly error budget** (≤ 24 , lower is better), and deliverable-accuracy reductions run against a **pinned baseline** the server enforces.

Forms that match how each KPI is measured

Employees report progress in **Activity Capture**. The form adapts to the KPI's measurement mode, so an SLA KPI asks for items-within-threshold, a ratio KPI asks for achieved-over-planned, and nobody ever types a percentage by hand. Saving an activity recomputes the KPI's provisional measurement immediately.

1
Pick the KPI
 Only the employee's own assignments are offered.

2
Pick the period
 The tracking period being reported, e.g. August 2026.

3
Enter the inputs
 The form asks exactly what the measurement mode needs.

4
Save
 Autosaved, duplicate-warned, and the measurement recomputes.

Measurement modes

Mode	What the employee enters	Example KPI
Ratio	Numerator (achieved) and denominator (planned / total)	Integrations vs planned · QA coverage (corrected ÷ identified)
Duration / SLA	Items inside the threshold and total eligible items	Requests turned around within 24 h / 2 business days
Count	The count achieved for the period	Safety hazard reports · monthly error budget (≤ 24 , lower is better)
Reduction	The current value — the baseline is pinned and enforced server-side	Reduce digitisation errors by 20% vs the pinned baseline
Milestone / Binary	Stage reached, or done / not done	Project phase delivery
Rubric	Score and maximum against an approved rubric	Innovation initiative assessment
Composite	Inputs of each component, weighted together	Database integrity (multi-check)

Every activity keeps its title, notes, reporter, **work date** and timestamp — the measurement is an *aggregate of captured work*, not a typed-in result. Daily and weekly KPIs are captured **incrementally**: batch logs accumulate through the period (QA coverage grows entry by entry), the form shows the running score and a cadence breakdown as work lands, and duplicate summary entries trigger a warning before anything is written. The activity feed gives managers a live view of updates as they arrive.

No official score without approved evidence

Results are claims; evidence makes them scores. From a KPI's detail page (or the Evidence Centre) an employee uploads files — documents, screenshots, exports, up to 25 MB — or attaches links, categorises them and ties them to the measurement they support. Files are stored in Convex storage and streamed back only through an **authenticated, authorization-checked** endpoint: a URL alone never opens someone else's restricted evidence.

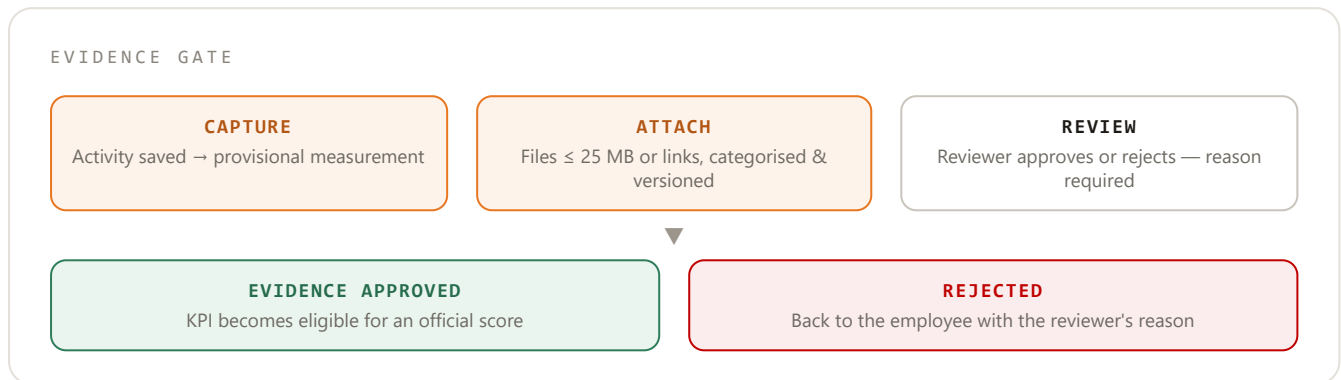


FIG. 2 — The evidence gate. A provisional measurement can exist without evidence, but a KPI whose definition requires evidence cannot be approved into an official score until a reviewer has approved that evidence.

- **Versioning** — replacing a file keeps the old version and its review history; nothing is overwritten in place.
- **Lineage** — every evidence item shows what measurement and period it supports, and every approved score can list the evidence behind it.
- **Restricted access** — employees see their own evidence; managers and reviewers see their scope; auditors see metadata without file contents; every download of a restricted file is logged.
- **Fast-track review** — reviewers approve evidence inline from the queue, with per-KPI bulk approve; every item carries a three-state badge (pending · approved · rejected), and an admin submission delete sweeps the KPI's attached evidence with it.

Rejection is feedback, not punishment. A rejected item carries the reviewer's reason back to the employee, who can attach a corrected version — the loop continues until the evidence stands or the claim is withdrawn.

Approval freezes a score you can reproduce

The **Review & Approval queue** lists provisional measurements grouped by employee and period, each with readiness flags: is required evidence approved? is any data-quality issue blocking the KPI? Reviewers work side-by-side — the claim, the activity behind it, the evidence — and verify or reject with a reason.

1

Provisional

Measurements recompute live as activity lands.

2

Verify

Reviewer checks evidence and recommends.

3

Approve period

Enabled only when every KPI is ready.

4

Snapshot

A reproducible score snapshot is frozen.

Approve period is a gate, not a button. It stays disabled until every KPI in the group has its required evidence approved and no blocking data-quality issue. Approving finalizes the measurements and freezes a **score snapshot** — calculation version, inputs, caps and weights — so the same number can be recomputed and audited months later.

Overrides & recalls — allowed, but never quiet

A manager or system admin can override a score — but the override **requires a written reason**, keeps the engine's original number alongside the overridden one, and writes both to the audit log. Rejections and period approvals can be recalled in the moment via a **post-action Undo**; a recall is just another audited decision. Employees see their own approval outcomes on their performance page — nothing about a decision is hidden from the person it concerns.

Locking a period

Once approvals are complete, the period can be **locked**: no further activity, evidence or overrides may touch it. Reopening a locked period is a system-admin action with a reason, and is itself audit-logged. Deadlines and period boundaries follow Africa/Lagos time.

Why it matters. The approval workflow is what turns the dashboard from a tracker into a system of record: a provisional number is a conversation; an approved snapshot is a commitment with evidence attached.

The engine scores; the AI only explains

Every score is produced by a deterministic engine — pure functions over the measurement inputs, the KPI's target, its direction of improvement and its weight. The same inputs always produce the same score, on any machine, with no model in the loop.

```
// per KPI, per period
attainment          = f(mode, inputs, target, direction) // 1.0 = 100%
cappedAttainment    = clamp(attainment, 0, officialCap)
weightedContribution = cappedAttainment × weightPoints
// employee score = Σ weightedContribution over all KPIs – out of 100 (80 core + 20 non-core)
```

- **Direction is configuration, not guesswork** — *higher-is-better* or *lower-is-better* is stored on each definition and never inferred at runtime.
- **Caps are policy** — attainment is capped at the official cap (default 100%), with a separate stretch cap (default 120%) available where the year's policy allows it.
- **Weights are points** — a 20-point KPI at 90% attainment contributes 18 points; the employee's score is the sum over all their KPIs, core and non-core labelled apart.
- **Honest edge cases** — percent-encoded reduction targets self-normalize; a zero baseline with zero errors counts as met; and a provisional measurement whose activities are all deleted is removed rather than scored as a fake 0% or 100%.

Status bands

Attainment maps to admin-editable status bands used consistently across every view, chart and export:

On / Above Target

Watch

At Risk

Critical

No Data

The LLM never scores. The optional AI narrative in reports (\$10) is generated *from* the engine's numbers and is labelled as explanation. No official score, attainment or band is ever produced, adjusted or rounded by a language model.

Reproducibility. Approved snapshots store the calculation version with the inputs — if the engine's rules ever change, old scores still recompute under the rules they were approved with.

From one employee's KPI to the whole team's trend

Oversight runs at three altitudes, each a click apart and each carrying the context that makes comparison fair:

- **Executive Overview** — the approved team score out of 100 (core and non-core labelled apart), submission-workflow posture, evidence and cadence coverage, overdue periods, the review backlog and the current score drivers.
- **Team Performance** — sortable scorecards with role and location comparison. Every comparison shows job role and configured weight, so a Lead and a Specialist are never ranked as if they carried the same targets.
- **Individual Performance** — one employee's KPIs with target, actual, attainment, weight and contribution; their activity timeline, evidence gallery, score history and report shortcuts.

Analytics workbench

The Analytics page adds coverage gauges, donut and comparison charts, data-quality exception views — and a **per-employee analytics** section with peer context: managers pick any employee in scope and see their attainment alongside the peer distribution, while an employee opening the same page sees their own numbers in context. Every chart offers a **table alternative** for accessibility, and filters are URL-encoded so any view can be shared as a link.

Live activity feed

Managers and admins follow a live feed of KPI updates as employees save activity — who reported what, against which KPI, moments after it happens. The same event stream powers in-app notifications and the email alerts described in §11.

Fair-comparison rule. Nowhere does the dashboard rank people on raw numbers alone — role, weight and scope ride along with every comparison surface, and provisional figures are always labelled as provisional.

Deterministic documents, provenance included

The Reports page runs export jobs with visible status and progress, preview, versioning and access logs. Two formats ship today, both rendered server-side in the Node runtime:

- **Excel workbook** — seven typed sheets (summary, per-employee scores, measurements, evidence index, data-quality ledger and more) with real cell types, formula-injection-safe text and the configured-weight note on every relevant sheet.
- **PDF report** — a deterministic layout of the approved numbers with **embedded engine-drawn charts** (weighted-score and per-KPI attainment bars, coloured by status band), generated from the same frozen snapshots the dashboard shows.

The optional AI narrative

A report can include a written narrative generated through the Vercel AI Gateway with structured output. The narrative **only explains numbers the engine already produced** — it summarises drivers, flags risks and drafts commentary — and it ships with a disclaimer and a human-approval status. If the AI key is absent, reports simply generate without the narrative; nothing else degrades.

Provenance on every document. Each generated report records who ran it, the format, the AI provider, model, prompt and schema version (when a narrative was included), and token usage — all written to the audit log. Report downloads are themselves access-logged.

Evidence-grade exports

Official scores come from frozen snapshots, so a re-run export reproduces the same numbers — a document derived from the system of record.

Anomalies become decisions, decisions become records

The workbook the catalogue was born from is imperfect — and the dashboard treats every imperfection as a first-class object. Import validation raises a **data-quality issue** for each anomaly: a blank frequency, a business unit spelled two ways, a unit mismatch between target and formula, a truncated formula cell.

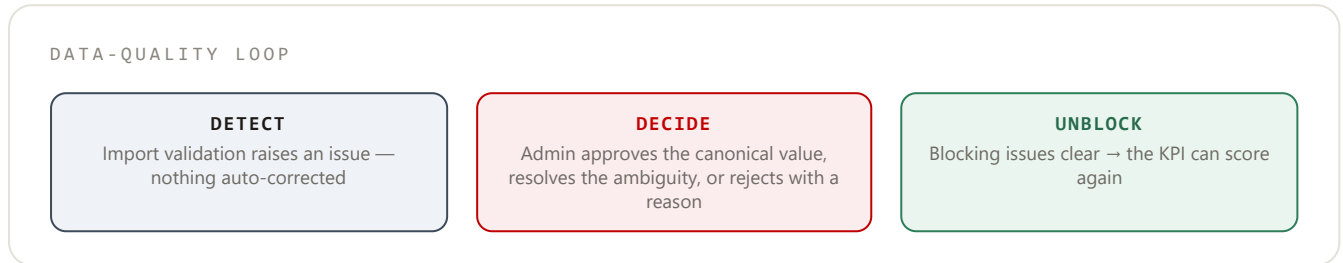


FIG. 3 — The data-quality loop. Source values are preserved verbatim either way, and a KPI with a blocking issue stays **scoring blocked** until a human decides.

The audit log

Every consequential action — configuration changes, imports, activity submissions, evidence reviews, approvals, overrides, report generation and downloads — writes an audit row: who, what, before/after, reason and timestamp, filterable by entity. Auditors and system admins read it; nobody edits it.

Notifications & scheduled jobs

- **In-app + email** — KPI updates notify the right people in-app and by email; if the email service is unconfigured the system degrades gracefully and the in-app trail still lands.
- **Daily overdue flagging** — a scheduled job flags tracking periods past their deadline (Africa/Lagos) so nothing quietly slips.
- **Review-backlog reminders** — managers with waiting approvals get nudged until the queue is clear.

Nothing moves silently. A number can change only through captured activity, a human decision, or a versioned configuration change — and each of the three leaves a record the audit log can replay.

From a day's work to a defensible score

Put together, the workflow reads as one unbroken chain: an employee captures real activity → the engine computes a provisional measurement → evidence is attached and reviewed → the period is approved and a snapshot frozen → analytics, reports and exports all read from that same snapshot → and every step along the way is attributable, timestamped and audit-logged.

● MANAGEMENT DASHBOARD

gis-kpi.vercel.app

Overview · Team · Review · Analytics · Reports · Data
Quality · Audit

● EMPLOYEE WORKSPACE

gis-kpi.vercel.app/activities

Activity Capture · Evidence · Individual Performance ·
Profile

Evidence-backed

Deterministically scored

Audited & exportable

The dashboard's promise is simple: **every score is reproducible from source activity and evidence**, every normalization of the source workbook is a surfaced, approvable decision, and nothing about anyone's performance is ever computed — or changed — in the dark.



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GIS KPI DASHBOARD

APPLICATION & PERFORMANCE WORKFLOW · V2.0

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